

Week 2: Authentication, Authorization and Incident Management

Project: RESQAI — AI-Assisted Humanitarian Disaster Message Classification

Authentication System

The second week focused on implementing secure user authentication. The system uses JWT-based authentication and password hashing with bcrypt. Three user roles were defined: Reporter, Admin, and Rescue Team. Reporters can submit and view their own incidents; admins can view all incidents and assign rescue teams; rescue teams can view assigned incidents and update their status.

Role-Based Access Control

Role middleware was introduced to ensure that users can access only the operations permitted for their role. This separates reporter, administrative, and rescue-team functionality and helps prevent unauthorized access.

Incident Management

The incident-management system was designed to store the message, reporter, location, classification category, confidence, urgency, priority, status, assigned rescue team, notes, and status history. Location information includes latitude, longitude, and address.

Incident Status

The incident lifecycle includes pending, assigned, accepted, en_route, on_site, and resolved. Status history with timestamps supports tracking the progress of each incident from submission to resolution.

Outcome

By the end of Week 2, authentication, authorization, and the core incident-management workflow had been implemented.